

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Business Administration (MBA)

Semester – II University Examination May 2010

MB 732 Financial Management

Date: 19.05.2010, Wednesday

Time: 12:30p.m. to 3:30p.m.

Maximum Marks: 70

Q-1 a How does the modern financial manager differ from the traditional financial manager? Does the modern financial manager's role differ; for the large diversified firm and the small to medium sized firm? **05**

Q-1 b What methods do you suggest for estimating working capital needs? Illustrate your answer with the help of following data: **05**

Material cost		Total product cost	968,000
Raw material consumed	248,000	Annual sales	1,448,000
Less: By-product	68,800	PBIT	480,000
Net material cost	179,200	Investment	1,600,000
Manufacturing cost		Period	1 year
Labour	171,200	Plant Life	10 years
Maintenance	160,000	PBDIT	640,000
Power and fuel	57,600	ROI (PBIT / (investment-depreciation)	33.3%
Factory overheads	240,000		
Depreciation	160,000		
Total manufacturing cost	788,800		

Assumptions:

Inventory: one month's supply of each of raw material, semi finished goods and finished material. Debtors: one month's sales. Operating cash: one month's total cost

Q-2 a What is beta? How is it measured? What are the problems in beta estimation? **05**

Q-2 b What are the pros and cons of debentures from the company's and investors' point of views? **05**

Q-3 a A company is currently paying a dividend of Rs 2.00 per share. The dividend is expected to grow at a 15% annual rate for three years, then at 10% rate for the next three years, after which it is expected to grow at a 5% rate forever. **05**

(a) What is the present value of the share if the capitalization rate is 9%?

(b) If the share is held for three years, what shall be its present value?

Q-3 b Your father has promised to give you Rs 100,000 in cash on your 25th birthday. Today is your 16th birthday. He wants to know two things: **05**

(a) If he decides to make annual payments into a fund after one year, how much will each have to be if the fund pays 8%?

(b) If he decides to invest a lump sum in the account after one year and let it compound annually, how much will the lump sum be?

(c) If in (a) the payments are made in the beginning of the year, how much will be the value of annuity?

Q-4 a A proforma cost sheet of a company provides the following data:

	Rs
Costs (per unit):	52.0
Raw materials	19.5
Direct labour	39.0
Overheads	110.5
Total cost (per unit)	19.5
Profit	130.0
Selling price	

The following is the additional information available;

Average raw material in stock: one month; average materials in process: half a month. Finished goods in stores: one month; Credit allowed by suppliers: one month; credit allowed to debtors: two months. Time lag in payment of wages: one and a half weeks. Overheads: one month. One-fourth of sales are on cash basis. Cash balance is expected to be Rs 120,000.

You are required to prepare a statement showing the working capital needed to finance a level of activity of 70,000 units of output. You may assume that production is carried on evenly throughout the year and wages and overheads accrue similarly.

Q-4 b An investor holds two equity shares x and y in equal proportion with the following risk and return characteristics:

$$E(R_x) = 24\%; E(R_y) = 19\%; \sigma_x = 28\%; \sigma_y = 23\%$$

The returns of these securities have a positive correlation of 0.6. You are required to calculate the portfolio return and risk. Further, suppose that the investor wants to reduce the portfolio risk (σ_p) to 15%. How much should the correlation coefficient be to bring the portfolio risk to the desired level?

Q-5a Two companies-Alpha Ltd. and Beta Ltd. are in the same industry with identical earnings per share for the last five years. The Alpha Ltd. has a policy of paying 40% of earnings as dividends, while the Beta Ltd. pays a constant amount of dividend per share. There is disparity between the market prices of the shares of the two companies. The price of Alpha's share is generally lower than that of Beta, even though in some years Alpha paid more dividends than Beta. The data on earnings, dividends and market price for the two companies are as under:

Alpha Ltd			
Year	EPS Rs.	DPS Rs.	Market Price Rs.
2000	4.00	1.60	12.00
2001	1.50	0.60	8.50
2002	5.00	2.00	13.50
2003	4.00	1.60	11.50
2004	8.00	3.20	14.50

Beta Ltd			
Year	EPS Rs.	DPS Rs.	Market Price Rs.
2000	4.00	1.80	13.50
2001	1.50	1.80	12.50
2002	5.00	1.80	12.50
2003	4.00	1.80	12.50
2004	8.00	1.80	15.00

Calculate (a) payout ratio, (b) dividend yield and (c) earning yield for both the companies.

Q-5b Firms A and B are similar except that A is unlevered, while B has Rs 200,000 of 5% debentures outstanding. Assuming that the tax rate is 40%; NOI is Rs 40,000 and the cost of equity is 10%. (i) Calculate the value of the firms, if the M-M assumptions are met. (ii) Suppose $V_b = \text{Rs}360,000$. According to M-M, do these

represent equilibrium values? How will equilibrium be set? Explain.

Q-6

Case: Pure Drinks

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Pure drinks is in the business of packaged fruit juices. It already produces and sells orange, apple and pineapple juices. It intends to increase its range of juices. The company engaged a well-known marketing consultant to conduct a market survey to find the scope for guava juice. The market survey indicated tremendous scope for guava juice, and that the company could sell 900,000 sachets of guava juice each year. The Rs. 5 million cost of the survey has not been yet paid in cash.

The survey has also shown that for one 1-litre sachet of guava juice, consumers are not prepared to pay more than Rs. 65. The management accounting department of the company has estimated that the variable cost per sachet would be Rs. 15, which includes material cost of Rs. 6.50, labour cost of Rs. 3.50 and overhead cost of Rs. 5.00. At a capacity of one million sachets, the fixed overhead cost per sachet is estimated at Rs. 6.75. The fixed overhead costs are allocations of the corporate general and administrative costs. Because of the project, the corporate marketing expenses and some administrative cost are likely to increase by Rs. 0.50 million per year. The management accountant feels that the profitability of the project would be affected by inflation. He expects 4% annual consumer price inflation and estimates that the price increases in variable overhead cost and labour cost would be 5% per annum and in material cost 3% per annum. The management is confident that they would be able to increase selling price by the expected consumer price inflation. The company is implementing a computer-based inventory control system that is likely to decrease the project's working capital by about 4% per year by sales volume. In the absence of the new inventory control system the working capital ratio is expected to be 30%. The working capital investment precedes sales by one year.

The company is considering using its existing building for processing the guava juice. The building was fully depreciated for tax purposes and was not in use so far. But a few months ago it received an offer of Rs 100 million (net of taxes) for the purchase of the building from a computer company. If the company does not sell the building immediately, its value is likely to appreciate by Rs 100 million (net of taxes) after five years. The processing facility for guava juice would cost Rs 200 million. It is expected to have an economic life of five years. The estimated salvage value of the processing equipment is Rs 100 million. The company uses straight-line depreciation method for tax purpose and tax is applicable on the profit or loss from the sale of the assets at the corporate income tax rate of 35%.

The company has no debt. The project will be financed from the internally generated funds. The shareholders expect a real rate of 8.65% from the proposed guava project.

Discussion Questions

1. If we ignore inflation, should the project be accepted? Explicitly state your assumptions and show calculations.
2. What shall be your decision if inflation is considered? Explain how inflation affects business decisions?